BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals

Research Tip: *Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.*

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

Database is an organized collection of data that is stored electronically and can be easily accessed, managed and updated

Databases are essential in the Modern world because are basically the backbone of almost every system we use today. Instead of searching through files manually data-base allow
- Instant queries
- Real-time results

Real-world example
- Banking system
- E-commerce (online shopping)

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

Data~~ware~~house is a large system used to store, organize, and analyze historical data from multiple sources for reporting and decision-making

*Purpose of Regular Database - Run daily operations
*Pursose of the Data warehouse - Analyze data
* Speed focus of regular database - fast inserts/updated
* speed focus of data~~ware~~house - fast read/queries
companies use weredate - Lase because managers can see trends compare performance, forecast future outcomes.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

- A data lake is a system where you store all your data in its raw, original format - without structuring or cleaning it first.

Concept a data lake is a centralized storage system that keeps all types of data in the raw original format
core idea it store first and structure lake. it comes from many sources. API, CSV, Databases
*A Business prefers a data lake when it needs flexibility, raw data storage and future analysis that isn't fully defined yet

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

A data mart is a smaller, focused version of a data warehouse that is build for a specific department or business function

Relationship
A data mart is a subset of that warehouse, filtered for a specific purpose
A data warehouse stores company-wide, cleaned data
*it is used by Business analytics, Department manager.

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

*Structured data define a highly organized in table with rows an columns
*Semi structured - does not fit in to tables, unstructured data is Data with no Predefined structure at all

Example
structured ID Name Amount
1 Kosebo 500

semi-structured JSON "name": John
"amount": 500

Unstructured data
PDF
image
videos

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

Data Modelling ~~is~~ is the process of designing how data structured, stored, and related in a System
The data modelling is preventing problems. · No duplicate data
· No missing relationship

Types of models
① Conceptual model · Big picture
· No technical detail
② logical model · more detail and Define entities and relationship

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action. Data Mining is the process of analyzing large dataset to to discover patterns, trends, and useful insights
* Retail - Retail companies analyze customer purchases to find buying pattern
* SMS System - Telecom companies analyze sms/call to to detect pattern and issues.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

Database management system is software that helps to store, manage, retrieve, and manipulate data in a database
* Oracle - handle millions of transactions per second
* PostgreSQL - Handle complex queries effeciently
* MySQL - work perfectly with PHP/Laravel
* MongoDB - you can choose MongoDB when you need
- Flexible Schema
large volume of semi-structured data

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics. ETL it's the process of collecting data, cleaning it and storing it in system.
E = Extract - you pull data from different sources (CSV API)
T = Transform - you clean and prepare the data (fix format)
L = Load - you store the cleaned data (reporting table)

10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

CSV - comma - separated value
Plain text, raw and - column format
uses when - Data is small to medium

JSON - Javascript object Notation -
Flexible - key value format that supports nesting
use when working with APIS

Parquet (columnar format)
Binary, column-based storage optimized for analytics
use for Handling big Data

Avro
Binary row-based format with schema support
Use for Data pipeline/streaming

Section B: Artificial Intelligence & Machine Learning

Research Tip: *Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.*

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each. AI is the ability of a computer or machine to perform task that normally require human intelligence

1 Narrow AI (weak AI)

AI designed to perform one specific task only

Example Siri: it can Answer question but cannot run business

2 General AI (strong AI) example A robot that can work as a doctor AI than can perform any intellectual task that a human can do

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

Machine learning is a branch of AI where computer learn from data and improve automatically without being explicitly programmed It differ with Traditional Programming by writing logic manually, Machine learning you provide data to Answer.

① Supervised learning model learns from labelled data

② Unsupervised learning model learns from unlabeled data

3 Reinforcement learning model learns by trial and error using rewards and penalties

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work? Deep learning is a type of machine learning that uses neural networks with many layers to learn complex patterns from data

AI - Machine acting intelligently

ML - Machine learning from data

A neural network is a system inspired by human brain made up of layers of ~~connection~~ connected "neurons" that process data ① input example SMS Data (image)

② Processing. Each neuron receives input

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

NLP is a field of AI that enables computer to understand, interpret and respond to human language, it understand text (message email)

Processing speech (voice commands)

① Voice Assistants

② Chatbots & messaging

3 Language Translation

15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for. LLM is a type of AI model trained on massive amount of text data to understand and generate human-like language and standard machine learning trained for one specific task, uses smaller dataset

① ChatGPT used for learning and explanations

② Google Gemini.

· Search assistance

· content generation

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does. Generative AI is a type of AI that can create new content such: text, image

or high level training on large data eg books, website

① ChatGPT: Generate text, Answer questions

② ~~DALL~~ DALL·E - Generate images from text descriptions

③ Github copilot

- Generates code suggestions
- Helps developers write codes faster.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect? ~~Data~~ Training dataset is the collection of data used to teach an AI or machine learning model how to perform a task. The golden rule is Garbage In - Garbage out

Good data - correct predictions

wrong predictions

If my sms data labels are wrong model will misclassify messages

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision. Computer Vision is a field of AI that enable machine to see, analyze and understand images or videos, machine don't see like humans - they see numbers (pixels)

① Self-Driving cars

② Face recognitions

Section C: APIs, Tools & Developer Concepts

Research Tip: *Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.*

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems? API is a way for different software system to communicate with each other
Real world = Restaurant you customer = your application
Kitchen = Server / System (e.g. CJD API)
• waiter = API
Important: system integration
Automation
Real time data

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly? Is a unique secretcode used to identify and authenticate your application when it makes request
Purpose: it confirms who is making the request
stored: store keys in .env files
Unauthorized access
others can use your API

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method. REST API is a type of API that follows a set of rules for how systems communicate over the web using HTTP
Stand REpresentational State Transfer
GET - Retrieve data from the server
POST - send data to create something new
PUT - update existing data
DELETE - Delete data from the server.

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object. JSON is a lightweight format used to store and exchange data. commonly used in APIs for easy to read and write

```
{"status": "success",
"message": "sms delivered"
"number": "2783234567"
}
```

23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

Is ~~engi~~ the process of designing and refering input(prompt) to get the best possible out put from AI model

Importance
clear prompt - correct answers

Poor prompt Explain API - Problem - Too broad

improve version explain what an API is in simple terms using a real world analogy

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript. A MAP function is used to apply a function (operation) to every item in a collection and return a new collection with the results

Importance It loops through data

Python example

```
number = [1, 2, 3, 4]
result = list(map(lambda x: x*2, number))
print(result
```

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model. A cloud computing is the delivery of computing services (servers, storage, databases, software; ① IaaS - provide basic infrastructure (AWS)

② PaaS - Provide platform to build and run application (Heroku)

③ SaaS provide ready-to-use software over the internet (Googledocs)

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are. a VCS is a tool that helps you track changes in files over time, so you can go back to previous version

Git is a distributed version control system used to track changes in code and ~~control~~ collaborate with other (and every change is recorded)

Commit a snapshot of your project at a specific point in time

Branch separate version of your project

Used to work on changes without affecting the main version

27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists? A jupyter notebook is an interactive tool that you write code, run it and display results is used by Python

① Interactive analysis: Run code in small parts

2 Great for visualization display charts

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

Python is a high level programming language used to build applications, analyze data, automate tasks and develop AI system:

. Is easy to learn with simple syntax
. less code compared to other languages

① Pandas - works with data in tabe
② Numpy - handle numerical operations
③ Matplotlib - It create charts and graphs
④ Scikit-Learn - It build machine learning model

Section D: Data Ethics, Governance & Careers

Research Tip: *Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.*

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

Data privacy refers to the protection of personal and sensitive information from unauthorized access.
It protect individual. prevents identity theft
GDPR is a data protection law designed to protect personal data and privacy of individual
Applies to EU European Union EU countries

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic. Data Ethics refers to the principles and standard that guide how data is collected
unetolical use of data ① using personal data without consent
A company collects users' phone number or email
It violates privacy
② Biased AI decisions. An AI system used for hiring
It has unfair outcomes
③ Misuse of AI generate content
create fake news spread misinformation

31. What is Data Governance? Define it and explain the key components of a good data governance framework. Is a set of policies, processes and standards used to manage data property across an organization
Key component: Policies & standards
Rules for how data is handled

Data governance ensures that data is accurate
Secure, well-managed

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences. Bias in AI happens when AI system produces unfair, skewed, or discriminatory results because of the data it was trained on or how it was designed.
A model is biased when it favor certain groups over others
Real-world Example of AI Bias

Hiring AI system - A company uses AI to screen job application!
What will happen AI learn that this group =

Unfair - discrimination
· Qualify people are rejected

33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools. Data Analyst - explain what happened
Data Scientist - predict what will happen
Data Engineer - build data pipeline / infrastructure
AI Engineer - build AI systems / models

Role	Skills	Tools
Data Analyst	SQL, Excel	Excel, Power BI
Data Scientist	statistics, ML, Python	Python, pandas
Data Engineer	ETL, Database	sql, Python
AI engineer	Deep learning	Python APIs

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions. BI is the process of collecting, analyzing and visualizing data to help organization makes better decision
- Microsoft Power BI - connect to data (Excel, SQL, APIs
It helps business with visualizes key metrics
- Tableau - Create interactive and advanced visualization
It helps business to makes complex data easy to understand

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

The most interesting and ~~suprising~~ concept is generative AI
It suprising because it can write report automatically
From my work I deal with - SMS logs
· CSD errors
· Report

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

The difficult concept I found most difficult to understand was deep learning, especially how neural networks work
Resource AI tools like chatGPT

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

Data an AI will play a major role in ~~improving~~ improving efficiency, decision-making, and automation in both my career and daily life

impact of my career

Data-Driven decision making

Automation of task

AI will reduce manual work

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. chatgpt (2026) openAI: https://chat.openai.com
2. w3schools (2026) w3Schools: https://www.w3schools.com
3. GeeksforGeeks (2026) GeeksforGeeks: https://www.geeksforgeeks.org
4. Khan academy ~~computer~~ Computer science and data fundamental
5. Python (2026) Software foundation. Python documentation Python.org
6. Microsoft (2026) Power BI Microsoft learning
7. Amazon web services 2026 cloud computing AWS: https://aws.amazon.com
8.

Good luck! We believe in you.

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